Neural Networks and Deep Learning Lab-4

Pakalapati Naga Srinivasa Varma

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1 Introduction

This experiment studies how gradients behave during neural network training. The experiment analyzes gradient behavior for different activation functions using a two-layer neural network and later extends the analysis to deep networks. The moon dataset is used since it is non-linearly separable and requires non-linear learning.

2 Gradient Monitoring and Analysis

2.1 Two-Layered Network

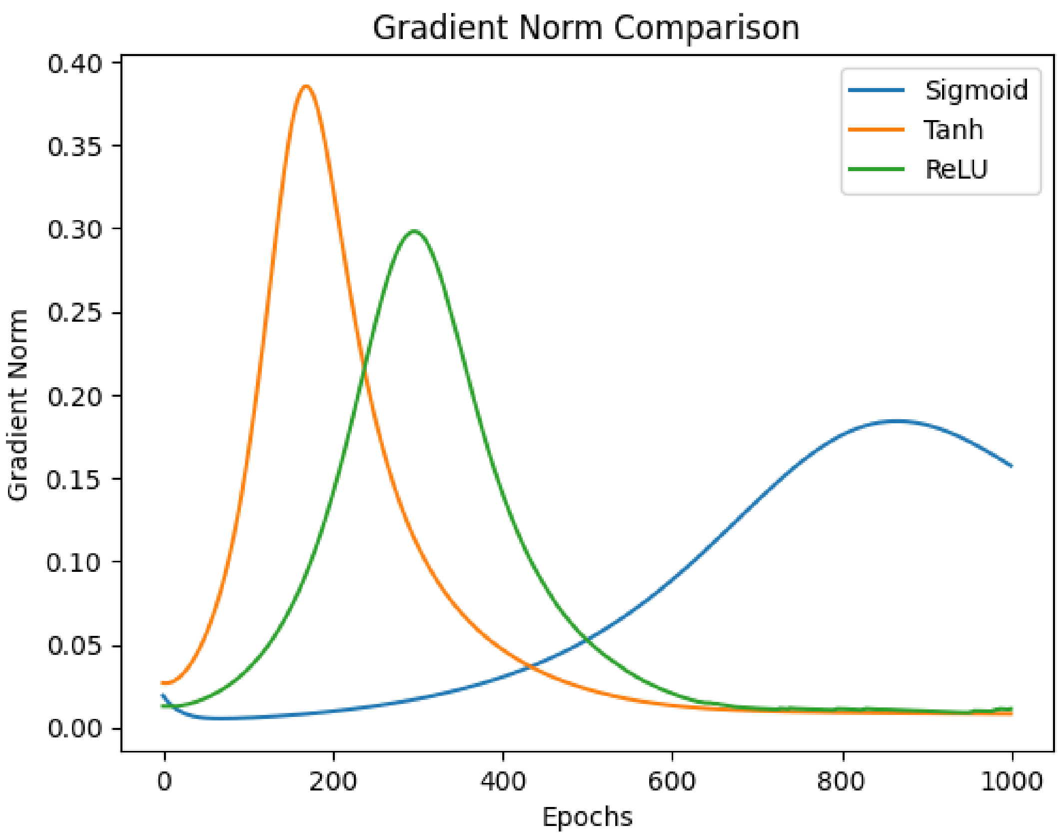


Figure 1: Gradients behavior for different activation functions

The above graph shows how gradient magnitude changes during training for different activation functions for the 2 Layer-Network.

2.1.1 Sigmoid Activation

- The gradient norm initially increases slightly as the network starts adjusting weights from random initialization. As training progresses, the gradient begins to decrease gradually.
- The sigmoid derivative is always small (maximum value of 0.25), which reduces gradient magnitude during backpropagation. Repeated multiplication of these small derivatives leads to the vanishing gradient problem, where earlier layers receive very small learning signals.

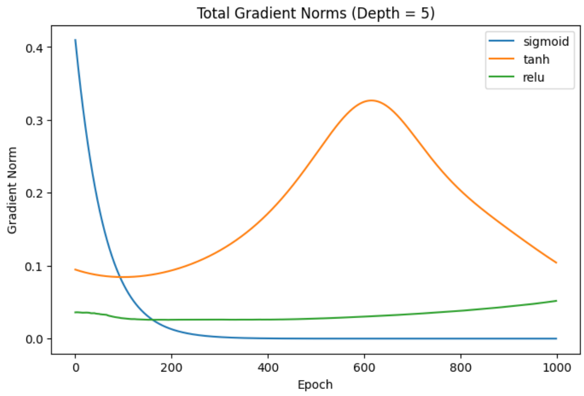
2.1.2 Tanh Activation

- Tanh produces stronger gradients near zero, which allows the network to adjust weights more aggressively at the beginning and reach the peak quickly.
- As training continues, neuron inputs become large and tanh outputs approach -1 or 1. In these regions, the tanh curve becomes flat, causing gradients to decrease rapidly.
- Although tanh delays vanishing gradients compared to sigmoid, it still suffers from vanishing gradient behavior in deeper networks.

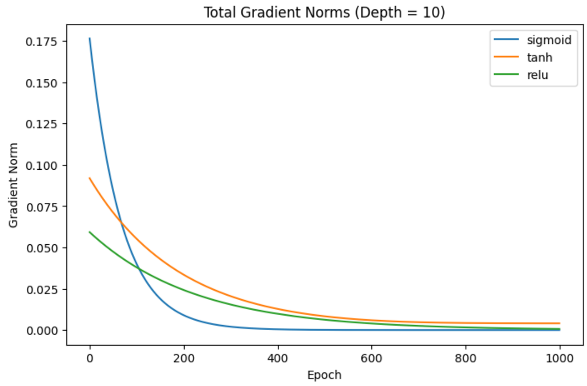
2.1.3 ReLU Activation

- ReLU produces strong gradients for neurons with positive inputs, allowing gradients to propagate without significant reduction. Unlike sigmoid and tanh, ReLU does not saturate for positive inputs, which helps reduce the vanishing gradient problem.
- Some neurons may produce zero gradients when their inputs are negative, but active neurons still allow effective learning. The gradient decreases gradually as training progresses mainly due to network convergence rather than gradient vanishing.

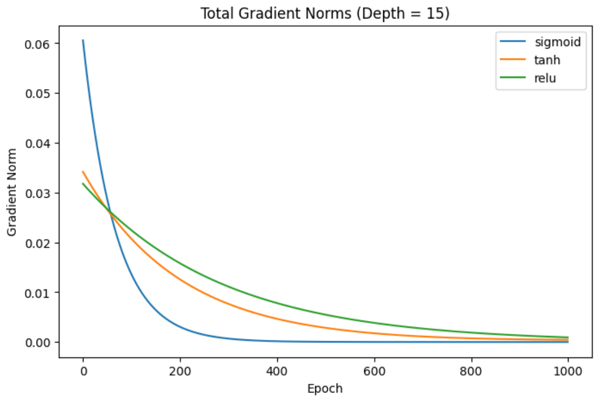
2.2 Deep Neural Networks



(a) Depth = 5



(b) Depth = 10



(c) Depth = 15

Figure 2: Gradient norms for deep neural networks with varying depths

2.2.1 Sigmoid Activation

- The gradient norm starts relatively high because neuron inputs remain near zero during initialization, where sigmoid produces its maximum derivative.
- In deep networks, gradients pass through multiple layers, and since sigmoid derivatives are always small, repeated multiplication causes rapid gradient decay.
- Compared to shallow networks, earlier layers in deep networks receive significantly weaker learning signals.
- As network depth increases from 5 to 15 layers, gradients vanish progressively earlier in training, demonstrating severe vanishing gradient behavior.

2.2.2 Tanh Activation

- Tanh produces stronger gradients near zero, allowing gradients to propagate more effectively across layers compared to sigmoid.
- Although tanh delays gradient decay, saturation occurs when neuron outputs approach -1 or 1, which reduces gradient magnitude.

- In deeper networks, gradients shrink consistently across training epochs due to repeated derivative multiplication.
- As depth increases, tanh shows moderate vanishing gradient effects, performing better than sigmoid but still losing gradient strength in very deep architectures.

2.2.3 ReLU Activation

- Unlike sigmoid and tanh, ReLU does not suffer from saturation for positive activations, which helps maintain learning signals even in deep architectures.
- As network depth increases, ReLU maintains significantly stronger gradients compared to sigmoid and tanh, demonstrating its suitability for training deep neural networks.
- The gradual gradient reduction observed with depth is mainly due to network convergence and inactive neurons rather than activation saturation.

2.3 Exploding Gradients

- Exploding gradients occur when gradient values grow excessively large during backpropagation, leading to very large weight updates and unstable training.
- This problem is more likely in very deep neural networks where gradients are repeatedly multiplied across multiple layers, causing exponential growth in gradient magnitude.
- Exploding gradients can also arise from large weight initialization or very high learning rates, both of which amplify gradient values and prevent proper convergence.
- They are commonly observed in recurrent neural networks where gradients propagate across long sequences of time steps, resulting in repeated gradient multiplication.
- In this experiment, exploding gradients were not observed because gradient norms remained stable across all activation functions and depths due to small weight initialization and bounded activation derivatives.

3 Conclusion

This experiment analyzed gradient behavior in shallow and deep neural networks using different activation functions. The results showed that sigmoid activation suffers from severe vanishing gradient problems, especially in deeper networks. Tanh provided improved gradient flow compared to sigmoid but still exhibited gradient reduction with increasing depth. ReLU demonstrated the most stable gradient propagation, making it more suitable for deep architectures.

The study also showed that vanishing gradients worsen as network depth increases, confirming theoretical expectations. Although exploding gradients are known to occur under certain conditions, they were not observed in this experiment due to controlled weight initialization and bounded activation derivatives. Overall, the experiment highlights the importance of activation function selection and proper initialization techniques for stable and efficient neural network training.